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How urban air quality affects land values: Exploring non-linear and threshold mechanism using explainable artificial intelligence

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Abstract:

Air pollution has a significant impact on human health and influences housing choices. Existing studies on determinant factors affecting property prices, including air pollution, mainly employed hedonic and spatial regression models that have limitations in capturing non-linear relationships and local interactions. Neglecting the non-linear relationships can lead to incomplete estimation impacts between explainable features and prices. To fill these gaps, this study used machine learning algorithms and SHAP (SHapley Additive exPlanations) to identify the relative importance of air pollutant variables on land values and their non-linear mechanism. The results showed that the Extreme Gradient Boosting (XGBoost) outperformed Ordinary Least Squares (OLS), Gradient Boosting Decision Tree (GBDT), Random Forest (RF), K-Nearest Neighbors (K-NN), and Extra Tree Regression (ETR) models in price prediction and capturing the non-linear relationship between air pollution and land values. Notably, Ozone (O₃) and Nitric Oxide (NO) concentrations contribute 1.64% and 1.47% to land value variation, respectively. The results

confirmed the non-linear relationship and identified that the threshold effect between air pollution and land values is at mean concentration values. Furthermore, we observed that the negative influence of O₃ and NO on land values appears as their concentration level is higher than 29.3 and 13 ppb, respectively. This paper contributes valuable insights by advancing our understanding of the non-linear impact of air pollution on house and land values, informing environmental policymaking, and shedding light on housing decisions.

Keywords: air pollution, house prices, non-linear relationship, machine learning, New York City

1. Introduction

Air pollution is an environmental factor that significantly impacts people's well-being (Klæboe et al., 2000; Ambrey et al., 2014). Much evidence has proven its negative impact on health, such as mental health, asthma, respiratory diseases, morality and cardiovascular systems (Who, 2013; Li and Managi, 2022; Zhang et al., 2022c). Additionally, the adverse effects of air pollution are intensive during the Covid-19 pandemic. For instance, air pollution enlarges Covid incidence rates (Fiasca et al., 2020), causes depression and morbidity (Chen et al., 2022b), and increases the fatality rate due to Covid-19 infections (Ogen, 2020). To mitigate the harmful impact of air pollution, households would consider pollution factors when making residential choices. House/land values, as reflected by a spatial equilibrium model, capture customers' preferences for an urban environment, which includes consideration of air quality (Lan et al., 2020). Generally, the influence of air pollution on land values can be explained by the people's willingness to pay for clean air. With the widespread dissemination of environmental information through various social media platforms, residents are increasingly attentive to air quality in their neighborhoods (Wang and Cai, 2021). Households are likely to be concerned about the air pollution level in the

residential location and tend to prefer a less polluted neighborhood during relocation. Therefore, improving air quality can positively impact the real estate market (Lai et al., 2021).

Many existing studies identified factors that affect land values. Previous studies used ordinary least squares (OLS) (Kang and Cervero, 2009; Michelson and Tully, 2018) and spatial regression models (Kim and Kim, 2016; Cohen et al., 2020) to explore the associations between built environment factors and land values. However, most studies focus on land characteristics and built environment features, while less explore the associations between air pollution and land values. Recently, mixed findings about the impact of air pollution on property values have been reported. Liang (2023) showed that PM_{2.5} concentration negatively impacts land values, and the prices to get cleaner air range from 32.08 to 33.13 USD/m². (Nishitateno and Burke, 2021) reported a negative influence of suspended particulate matter (SPM) on land values and the elasticity of land values with SPM concentration between -0.4 and -1.0. Most studies found negative impacts of air pollution (including SO₂, PM_{2.5}, O₃) on house/land values (Yusuf and Resosudarmo, 2009; Nishitateno and Burke, 2021; Amini et al., 2022; Wang et al., 2022b), while Mei et al. (2020) and Usharani (2021) claimed that house prices have an insignificant relationship with SO₂ concentration and that rising O₃ concentration is associated with increasing house/land values. Furthermore, studies mentioned above used linear methods to identify the relationship between pollution and land values/house prices, yet an increase in pollution concentration does not always result in a linear decrease/ increase in land values. Relying on linear assumptions may lead to overestimating/ underestimating the effects of air pollution on land values. As the influencing mechanism of air pollution on real estate values is less analyzed at the micro-scale, the threshold at which air pollution concentration starts to capitalize on property prices (i.e., land values) needs further investigation.

Machine learning techniques are widely used in classification and regression models and have better performance and advantages compared to conventional models, such as OLS and Geographical Weighted Regression, in land value prediction and non-linearity examination (Zhang and Haghani, 2015). For instance, tree-based algorithms are relatively robust to outliers and noise (Duerr et al., 2018), and random forest is effective in non-linear relationship exploration (Kiely and Bastian, 2020). Despite their outperforming performance, machine learning outcomes are difficult to interpret. Recently, techniques such as SHAP (SHapley Additive exPlanations) can be used to interpret machine learning models and quantify the contribution of explanatory variables, including non-linearity and interaction effects (Huo et al., 2022).

To fill the research gaps mentioned above, this study employed machine learning algorithms and SHAP to explore the complicated non-linear relationships between land values and air pollution. We used land values as a prediction variable (Wen and Goodman, 2013). First, we employed various machine learning models to select the optimal algorithms for estimating relationships between land values and air pollution. Second, we incorporated SHAP with a selected machine learning model to interpret results. Specifically, the aims of this work are to (1) identify the optimal machine learning algorithms to explore the relationship between land values and air pollution; (2) explore the relative contribution of air pollution features on land values; (3) identify the non-linear relationships between air pollution and land values at a micro level, which is missing in the body of literature; and (4) determine the threshold effects of air pollution on land values. This study contributes to the existing literature by examining non-linear mechanisms of air pollution on land values using explainable artificial intelligence algorithms. By identifying the thresholds at which air quality begins to negatively impact land values, policymakers can make

informed decisions regarding urban planning, pollution control measures and interventions in the housing market.

The remainder of this work is structured as follows. Section 2 provides literature reviews on land value studies and the relationships between air pollution and property prices. Section 3 proposes hypotheses. Section 4 presents the study area, data and methodology. Section 5 presents the comparison of the main results of machine learning performance and the non-linear relation between air pollution features and land values. Section 6 discusses the findings, and Section 7 concludes.

2. Literature review

2.1. Land value studies: from conventional to modern approaches

The relationship between house/ land values and urban features was explored in previous studies using both conventional and modern techniques. Conventional methods, such as hedonics, have been widely employed, while modern methods, especially machine learning techniques, have emerged as increasingly prevalent in recent years.

The core of traditional methods is evaluating prices by utilizing relevant property characteristics and their determinants (Wang et al., 2017b; Doan, 2023). They are effective techniques for determining the linear correlation between land values and their characteristics (Li et al., 2021; Zhang et al., 2023b), including urban environment (Geoghegan, 2002), traffic noise (Kim et al., 2007), accessibility (Wang et al., 2015; Yang et al., 2020), land use (Kang, 2019), urban park (Kim and Yoon, 2023), urban ecosystem (Chen et al., 2022c), and economy (Mirakatouli et al., 2018). For instance, Jim and Chen (2009) evaluated how the natural landscape influences property values using hedonic models. The results revealed that the harbour view increased property prices in Hong Kong. Zhang et al. (2022c) used OLS models to explore the association

between air quality and house prices at the provincial scale. The authors concluded that an increase of 1-unit PM_{10} concentration corresponds to a 0.24% decrease in house prices. Additionally, the relationship between apartment building prices and ozone concentration levels was confirmed by Hitaj et al. (2018), as air quality was capitalized in apartment building prices. However, these studies do not quantify the relative importance and to what extent air pollution capitalizes on prices.

As for machine learning methods, scholars have adopted several popular models for property price predictions and its determinants, such as random forest (Yoo et al., 2012; Zhang et al., 2021), support vector machine (Chen et al., 2017; Ho et al., 2021), gradient boosting decision tree (Kang et al., 2021; Iban, 2022), K-nearest neighbors (Ma et al., 2020), and eXtreme Gradient Boosting (Taecharungroj, 2021). Using XGBoost as a main method, Kim et al. (2021) proposed a framework to estimate land value appreciation rates with high accuracy. While the Quantile Random Forest is the best model in the research by Carranza et al. (2022), the random forest (RF) has impressive performance in land value prediction in the study of Ma et al. (2020). Additionally, Rico-Juan and Taltavull de La Paz (2021) provided evidence on the prediction capabilities of various machine learning methods and pointed out the superiority of RF in house price prediction with low error. Park and Bae (2015) proposed a prediction model by comparing machine learning algorithms, including C4.5, RIPPER, Naïve Bayesian and AdaBoost. Potrawa and Tetereva (2022) applied machine learning models to uncover the features and complicated relationships between window views and real estate prices that traditional methods cannot reveal. To sum up, machine learning techniques offer superior performance over conventional methods in comprehensively characterizing the complex relationships between land values and their features, encompassing both linear and non-linear associations. However, the relationships in machine learning are

difficult to interpret since they are considered black-box models. Thus, combining machine learning and interpretation algorithms is essential to gain insights into the underlying relationships and factors driving the model's predictions.

2.2. Air pollution and property prices

Previous studies have examined the linear impact of air pollution on real estate values (Liang, 2023). Chen and Chen (2017) assessed the influence of PM_{2.5} concentration on house values using panel data from 286 cities, concluding that real estate prices decreased by 46 RMB/m² as PM_{2.5} increased by 1 µg/m³. Chen and Jin (2019) proposed a strategy based on ventilation coefficients to explore the relationship between air pollution and housing prices at the city level. Ou et al. (2022) proved the negative association between house prices and PM_{2.5}, demonstrating that in cities with a population of 1 to 5 million, a 1% increase in PM_{2.5} is associated with a 0.32% decrease in house prices. Ou et al. (2022) measured the benefits of air quality capitalization enhancement in housing sales and rental markets over a period of 4 years across 283 Chinese cities, gaining results similar to the study of Amini et al. (2022). The findings above mostly demonstrate the linear relationship; nevertheless, non-linear relationships between urban objectives always exist, as the increase in explainable variables may not immediately lead to an increase in property prices. Assuming a linear relationship, escalating determinants would lead to proportionately inflated property prices, reaching implausibly high values and thus resulting in biased estimations. However, to the best of our knowledge, few scholars explored the non-linear relationship between property prices and air pollution. Only two papers on the Web of Science by Zou et al. (2022) and Zhang et al. (2023c) confirmed the non-linear relationships between housing values and air pollutants using GBDT. In addition, since most studies above are conducted at macro levels, such as the city and district scales, their ability to reveal the detailed influence mechanisms of air

pollutants on housing prices at micro scales is limited (Chen and Jin, 2019; Wang et al., 2022b), and cannot reveal parcel and block characteristics that are crucial for accurate predictions of property values.

In summary, studies concerning the non-linear impact of air pollution on property prices at the micro level are quite limited. Most studies confirmed their negative linear relationship at the macro level, such as the city and district scale. In addition, most scholars focused on using machine learning models to achieve better prediction performances compared to traditional ones. Additionally, due to its black-box nature, interpretations of the non-linear relationship in the model's output are lacking in previous studies on air pollution and house prices. Thus, the non-linear relationships and threshold effects at the micro level are still gaps in the body of literature.

3. Hypothesis

Air pollution remains a pressing environmental concern globally, prompting a heightened exploration of its repercussions on individuals living in polluted areas and the adverse health effects it may induce (Zhang et al., 2022b; Liang, 2023). Apart from well-being, air pollution also affects people's decision-making and behaviours, such as attitudes, physical activities and expenditure preferences (Ke et al., 2021; Wang et al., 2022a). Air quality in a place affects the accommodation willingness and residence choice, thus generating impacts on local land values (Chen and Jin, 2019). Driven by concerns for physical health and overall living quality, individuals naturally gravitate towards clean air. Residents are often willing to invest more in areas boasting good air quality, eschewing regions burdened by high pollution levels. Consequently, this study posits the following hypothesis:

H1: Air pollution negatively affects land values.

Drawing inspiration from the protection motivation theory proposed by Rogers (1975), this study delves into the behavioral dynamics driven by fear appraisals in the realm of pollution economics. At lower pollution concentrations, individuals may display a reduced concern for health impacts in settlement decisions, possibly prioritizing urban facilities in their purchase choices (Wang et al., 2022b). However, when air pollution surpasses a critical threshold, perceived health risks escalate, diminishing the allure of residing in such areas and, consequently, impacting land values. Individuals may perceive relocation to areas with lower pollution levels as a pragmatic response (Chen et al., 2022a), which in turn affects land values. This leads to the formulation of our second hypothesis:

H2: The effects of air pollution on land values following non-linear and threshold mechanisms.

4. Study area, data and methodology

4.1. Study area

New York City (NYC) in the northeastern region of the United States, is situated at the southeastern tip of New York State (Figure 1). The city comprises five boroughs: Manhattan, Brooklyn, Queens, the Bronx, and Staten Island.

Air pollution is a major environmental issue in many global cities, and NYC is no exception. NYC is a bustling metropolis with a high population density, numerous industries, and many kinds of vehicles, all contributing to the high levels of air pollution. With over 8 million people and numerous industries, the city produces a significant amount of air pollutants that negatively impact the environment. The primary sources of air pollution in New York City are transportation, power generation, and industrial activities (NYC Health, 2018). Additionally, natural disasters such as wildfires and dust storms also contribute to air pollution. As a result, New York ranks 14th most polluted city in the USA. Air pollution in NYC also has numerous negative

impacts on human health. For example, in New York City, the estimated annual number of asthma emergency visits due to ozone is 5115. Additionally, there were 410 cardiac deaths and 1971 respiratory deaths due to ozone and PM_{2.5}, respectively (Environment & Health Data Portal, 2022).

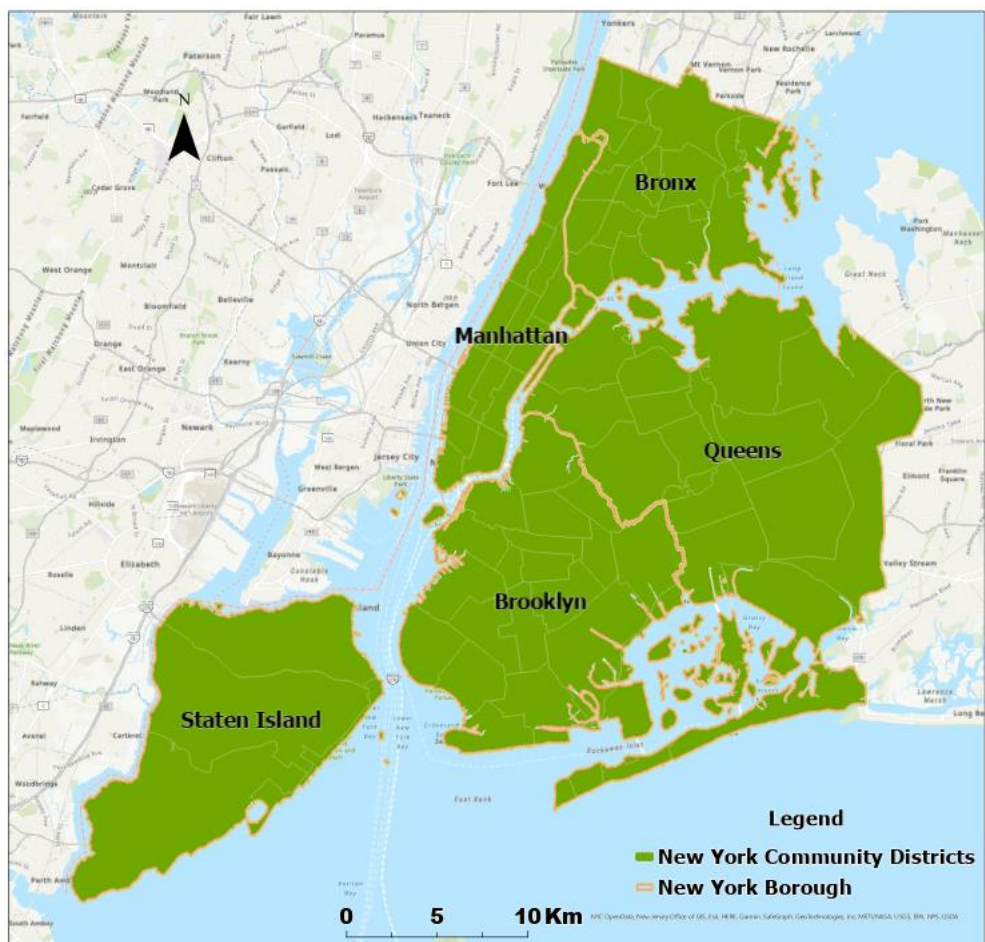


Figure 1. The location of New York City, USA

4.2. Data

This paper aims to identify the influence of pollutant characteristics on land values at the lots level. Specifically, we incorporate variables related to the lots' characteristics and surrounding built environment features which have been demonstrated to be influential in land values and house prices in previous studies (Wen et al., 2014; Wang et al., 2017b), and air pollution variables. We controlled potential socioeconomic factors by categorizing samples based on five distinct districts. The descriptive statistics of variables are shown below (Table 1).

The assessed land value for tax is the dependent variable, which is a part of MapPLUTO (Primary Land Use Tax Lot Output), collected from the NYC planning portal. The Digital Tax Map (DTM) from the Department of Finance is combined with PLUTO tax lot data in MapPLUTO. It also includes substantial geographic and land use data at the tax lot level in the ESRI shapefile and File Geodatabase formats. In this study, we only used the residential lots for analysis to maintain a more consistent and reliable dataset, reducing potential data discrepancies and enhancing the robustness of the findings. Additionally, residential areas are of particular interest due to the direct impact of air quality on residents' health, well-being, and quality of life. By narrowing our focus to residential land lots, we aim to provide a focused and in-depth analysis of how air pollution impacts the residential real estate market, considering its unique dynamics and residence preferences. After data cleaning, 5025 residential lots were extracted from MapPLUTO for official analysis. The number of multi-family walk-up and elevator buildings is 4055 and 970 records, respectively.

The POI data collected from Overpass Turbo provides various open data such as road networks, footprint buildings, and POI. 11 kinds of POI were used in this study, including education (school, college and university), health care services (hospital and health centers), market, business, theatre, park, restaurants, bank, convenience store, metro station, and department store. We computed the count of POI within a 500m radius of land lots. The selection of this distance is based on the consideration that it represents a convenient walking distance commonly used to delineate the boundaries of residential neighborhoods.

The air pollutant data was collected from NYC Open Data, which provides a citywide raster of annual average predicted surface levels for nitrogen dioxide (NO₂), fine particulate matter (PM_{2.5}), black carbon (BC), nitric oxide (NO), summer average for ozone (O₃) and winter average

for sulfur dioxide (SO₂). In this study, we used the pollutant data of BC, NO, NO₂, PM_{2.5} and O₃ of 2021. As meta-analyses have indicated that the spatial range of mobile sources is approximately 100–400 m for particulate matter and 200–500 m for NO₂ (Karner et al., 2010), we opted for a buffer size of 500 m. We computed each lot's average air pollution concentrations within this buffer area to reflect the ambient air pollution levels.

Table 1. Descriptions of variables

Label	Variable	Description	Mean	St. D	Min	Max
1	Price_per_foot	Price per square feet	262.423	335.111	7.318	4135.995
2	BldgArea	The total gross area, except condominium (square feet)	20670.854	63509.012	380	2.81E+12
3	ComArea	Area used for commercial purposes (square feet)	417.031	2577.723	0	72803
4	ResArea	Area used for residential purposes (square feet)	20280.088	63095.716	1000	2812639
5	OfficeArea	Area used for office purposes (square feet)	4.402	180.347	0	12075
6	RetailArea	Area used for retail purposes (square feet)	53.712	879.467	0	30324
7	GarageArea	Area used for garage purposes (square feet)	284.960	1988.769	0	53254
8	StrgeArea	Area used for storage/loft purposes (square feet)	7.834	110.708	0	3762
9	NumBldgs	The number of buildings	3.077	10.702	1	207
10	NumFloors	The number of floors	4.360	3.328	3	70
11	UnitsRes	The number of residential units	19.945	83.361	4	3857
12	UnitsTotal	The number of residential and non-residential units	21.912	84.708	4	3857
13	LotFront	The lot's frontage (feet)	48.715	81.435	20	3617
14	LotDepth	The lot's depth (feet)	95.395	116.103	89	5360
15	BldgFront	The building's frontage along the street (feet)	30.904	65.477	0	3705
16	BldgDepth	The building's depth (feet)	47.642	37.364	0	483
17	YearBuilt	The year of construction	49.472	45.731	1	231
18	BuiltFAR	The floor area ratio	2.537	1.625	1.71	23.13
19	ResidFAR	The residential floor area ratio	2.472	1.554	1.35	10
20	CommFAR	The commercial floor area ratio	0.292	1.169	0	15
21	FacilFAR	The facility floor area ratio	3.33	1.770	2	15
22	BoroCode	The borough where lots are located (1: Manhattan; 2: Bronx; 3: Brooklyn; 4: Queens; 5: Staten Island)	2.845	0.915	1	5
23	CondoNo	The number of condominiums assigned to complex	1979.774	1537.446	1	5599
24	Irrlotco	Shape of lot (1: irregular shaped; 0: not irregular shaped)	0.187	0.390	0	1

Label	Variable	Description	Mean	St. D	Min	Max
25	Landuse	Land use categories (1: Multi-Family Walk-Up Buildings; 2: Multi-Family Elevator Buildings)	1.193	0.394	1	2
26	Proxi_code	Physical relationship of the building to neighboring buildings (0: Not available; 1: Detached; 2: Semi-attached; 3: Attached)	0.483	1.018	0	3
27	Bsmt	The building basement	4.818	0.748	1	5
28	N_school	Number of schools within 500m	2.672	3.055	0	20
29	N_hospital	Number of hospitals within 500m	1.439	2.869	0	27
30	N_business	Number of businesses within 500m	58.773	60.871	0	520
31	N_market	Number of markets within 500m	2.234	1.900	0	16
32	N_theatre	Number of theatres within 500m	0.464	1.211	0	15
33	N_park	Number of parks within 500m	0.312	0.655	0	4
34	N_restaurant	Number of restaurants within 500m	28.339	33.787	0	235
35	N_bank	Number of banks within 500m	2.466	3.475	0	32
36	N_department	Number of departments within 500m	0.284	0.810	0	7
37	N_convinence	Number of convenience stores within 500m	6.715	7.106	0	35
38	HubDist	Distance to the city center (meters)	8340.638	4767.372	0	31523.210
39	Shortline	Near the shortline (1: Yes; 0: No)	0.443	0.496	0	1
40	BC	Black carbon concentration (ppb)	0.655	0.130	0.372	1.175
41	NO ₂	Annual average Nitrogen dioxide ($\mu\text{g}/\text{m}^3$)	17.996	2.200	8.428	26.393
42	NO	Annual average Nitric oxide (ppb)	13.187	4.147	5.979	40.020
43	PM _{2.5}	annual average fine particulate matter <2.5 microns ($\mu\text{g}/\text{m}^3$)	6.884	0.592	5.863	9.201
44	O ₃	Summer average ozone (ppb)	29.376	1.496	25.455	35.882

4.3. Methodology

4.3.1. Analytics framework

Figure 2 describes the four-step analytical framework. Firstly, land values, built environment, and air pollutant data are collected and cleaned. Secondly, potential factors are selected using the traditional regression model. We divide data into training and validation sets at a ratio of 70:30 before using five machine-learning models. The performances of machine learning models are compared, and the model with higher accuracy is selected for the following steps. Thirdly, the relative importance of explanatory variables is estimated using the most appropriate machine learning technique and SHAP summary plots. We then explore the non-linear relationship

between air pollutant variables and land values and the threshold impacts. Finally, we discuss the potential implications of air pollution for house price management and urban governance.

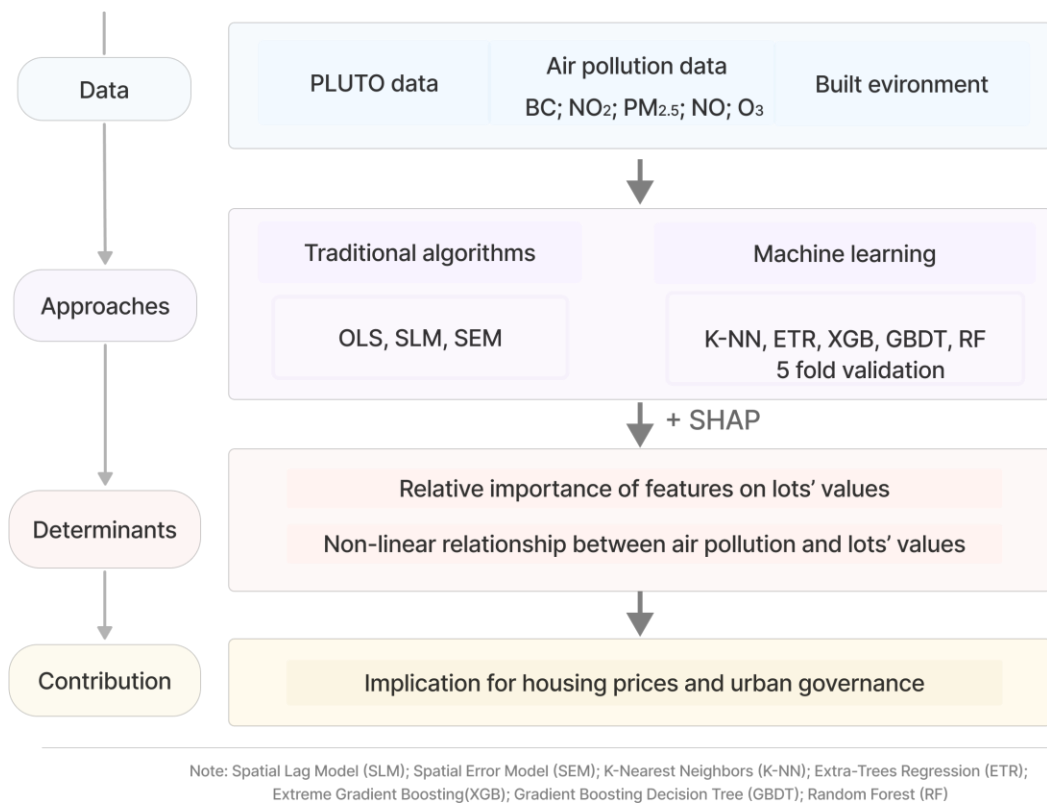


Figure 2. The analytical framework

4.3.2. Method

4.3.2.1. Hedonic regression method

Firstly, we employed the OLS model to select potential variables affecting land values. The OLS model is a basic statistical method to predict the dependent variable based on various independent variables (Hu et al., 2022; Doan, 2023). The hedonic model assumes a linear connection between the dependent variable and independent variables and seeks to minimize the sum of squared residuals. Considering that spatial autocorrelation may exist among the variables that the OLS model can not capture, we also adopt the spatial lag model (SLM) and spatial error model (SEM) to account for the spatial effects. The OLS (Eq. 1), SLM (Eq. 2) and SEM (Eq. 3 & 4) can be expressed as follows:

$$P = \alpha_0 + \sum_i^n \alpha_i . X_i + \varepsilon \quad (\text{Eq.1})$$

$$P = \rho W_y + X\alpha + \epsilon \quad (\text{Eq.2})$$

$$P = X\alpha + \theta \quad (\text{Eq.3})$$

$$\theta = \lambda W_\theta + \mu \quad (\text{Eq.4})$$

Where, P is land values; α_i is coefficient of dependent variables, X_i is the i th independent variables; and ε is residual error; W_y represents a weighted average of land values at nearby locations, and ρ is a spatial dependence parameter. θ is a vector of spatially autocorrelated residuals; λ is an autoregressive parameter; W_θ is a weighted average of error terms at nearby locations; μ is a vector of independent residuals.

4.3.2.2. Machine learning method

Performances in land values prediction of five machine learning models were compared, including K-Nearest Neighbor Regression (KNN), Gradient Boosting Decision Tree (GBDT), Extra Tree Regression (ETR), Extreme Gradient Boosting (XGBoost), and Random Forest (RF). The model with better performance reveals a more accurate depiction of the relationship.

(1) K-Nearest Neighbor Regression: This type of supervisor machine learning technique is effective for regression and classification. It learns by contrasting the provided test instance (A, b) with the training set (A_i, b_i). For example, KNN calculates the difference (d_i) between A and A_i before sorting the distances. If d_i ranks in i^{th} place, the instance related to d_i is the i^{th} nearest neighbor.

(2) Gradient Boosting Decision Tree is a regression model, an ensemble learning method based on Classification And Regression Trees (CART). GBST's trees are built to fit

the residual of trees at an iteration. The aim of GBDT is to choose a classification function to minimize the aggregation of the loss function.

(3) Extra Tree Regression is developed as an extension of the random forest algorithm, and it is less likely to overfit a dataset (Ahmad et al., 2018). Similar to RF, Extra Tree trains each base estimator using a random selection of characteristics. However, it randomly selects the best feature along with the corresponding value for splitting the node (Geurts et al., 2006).

(4) XGBoost is a machine learning method that conducts a boosting process to construct an accurate prediction model. XGBoost was proposed by Chen and Guestrin (2016) and was widely used for property price prediction. The core algorithm of XGBoost is gradient boosting, an ensemble technique that integrates weak learners to generate a stronger learner. Each new weak learner is designed to correct the deficiency of the previous learners. XGBoost can be used for both classification and regression purposes.

The objective of XGBoost contains of two parts, including training loss and regularization, as shown in Eq.5:

$$Obj(\varphi) = L(\varphi) + \Omega(\varphi) \quad (\text{Eq.5})$$

Where, L represents the training loss function and Ω represents the regularization. While $L(\varphi)$ estimate the model performance on training set, $\Omega(\varphi)$ has function to control the model complexity such as overfitting or underfitting. In addition, the complex of each tree is computed as in Eq. 6:

$$\Omega(f) = \sigma \cdot l + \frac{1}{2} \vartheta \cdot \sum_j^t \rho_j^2 \quad (\text{Eq.6})$$

Where, l is the number of leaves and ρ is the vector of scores on leaves. The structure score of the model is defined in the objective function as Eq. 7:

$$Obj_j = \sum_j^l [G_j \rho_j + \frac{1}{2} (H_i + \vartheta) \cdot \rho_j^2] + \sigma \cdot l \quad (\text{Eq.7})$$

Where, γ_j represents independent of each other. $G_j \rho_j + \frac{1}{2} (H_i + \vartheta) \cdot \rho_j^2$ is quadratic and the best ρ_j for a given structure $q_{(x)}$.

(5) Random Forest: This method was proposed in 2001 by Breiman (2001). Random Forest combines N_T CARTs and accumulates results by voting or averaging. Specifically, each CART is built based on a random selection of variables and samples that are selected by bootstrapping. For instance, a dataset is named $\text{Data} = (X, Y)$, and N_T sample sets are produced. Each set has $D_j = (X_j, Y_j)$ and accounts for 2/3 of the Data. The variable selection is identified by the Gini value as Eq. 8.

$$\text{Gini}(\text{Data}) = 1 - \sum_{i=1}^{\tau} P_i^2 \quad (\text{Eq. 8})$$

Where τ is the number of classes in the label and P_i is the probability of the presence of the i^{th} class. In case of CART is split according to variables B , the Gini values afterward is calculated as follows:

$$\text{Gini}(\text{Data}, B) = \sum_{i=1}^m \frac{|D^i|}{|D|} \cdot \text{Gini}(\text{Data}^i) \quad (\text{Eq 9})$$

Where m presents the number of branches when splitting based on attribute B . Generally, Random Forest is a highly scalable and parallelisable ensemble method. In the regression model, it trains a set of decision trees separately and then employs averaging to generate predictions.

Four types of accuracy indicators are adopted for performance evaluation: R-squared (R^2), Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

4.3.2.3. *Relatively importance of variables and non-linear relationship by SHAP*

For prediction of land values, explanatory variables rarely have the same correlation with the target variable, as the weights of each variable usually differ (Friedman and Meulman, 2003). Thus, identifying the contribution of each independent variable to the prediction variable provides insights into more accurate model prediction and decision-making.

The existing machine learning techniques are considered black boxes, so several methods are used to interpret the system's structure, such as Partial dependence plots (PDP), Accumulated Local Effect (ALE), and SHAP. Among them, SHAP provides a comprehensive analysis of global, local, and interaction effects, which cannot be effectively captured by PDP and ALE methods. In the SHAP model, the Shapley value presents the contribution level of explanatory variables to the prediction variable (Li, 2022). Thus, combining machine learning techniques and SHAP reveals the relative importance of variables, non-linear relationships, threshold, variables' interaction and synergies (Chen et al., 2023b). The contribution of each variable (ϑ_i) on the output model $\sigma_{(N)}$ is allocated based on marginal contribution. Eq 7 presents Shapley values.

$$\vartheta_i = \sum_{S \subseteq N} \frac{|S|! (M-|S|-1)!}{M!} [\varphi(S \cup i) - \varphi(S)] \quad (\text{Eq 7})$$

Where, S is a subset of features used in the model, M represents a number of the features, N represents the set of all input features. $\varphi(S \cup i) - \varphi(S)$ describes the deviation of Shapley values from their mean.

5. Result

5.1. Exploratory variables selection

Firstly, we employed a stepwise OLS regression model to examine the influence of variables on land values and selected variables with a significant level of confidence. Secondly, we tested multicollinear relationships between variables using the package “car” in RStudio. Variables that have a VIF value higher than 7.5 were eliminated. The Durbin-Watson value is 1.592 (acceptable range: 1.5 to 2.5). Seventeen variables remained in the model after OLS regression (Table 2). It is noticeable that O₃ and NO have a significant negative influence on land values. On average, as the concentration of NO and O₃ increased by 1ppb, the land values decreased by around \$6 and \$14/square foot, respectively. Overall, lots' internal characteristics positively impact land values, such as the number of buildings, the number of floors, the built floor area ratio, the residential floor area ratio, and the commercial floor area ratio. Furthermore, land values tend to decrease as the distance from the city center increases. The higher the number of surrounding theatres, restaurants, and banks, the higher land values are. By contrast, the number of condominiums, year of building, and land use negatively influence land values. Specifically, land values with multi-family walk-up buildings are \$126/ square foot lower than elevator buildings. Regarding geography, land values in Manhattan are higher than in the Bronx, Brooklyn, Queens, and Staten Island. The VIF of variables are comparatively small (below 7.5), indicating that multicollinearity does not affect the reliability of results. The coefficients of 17 variables are all statistically significant in the land value prediction (P-value <0.05).

Table 2: Estimate results of the Stepwise OLS model

Term	Estimate	Std. error	Statistic	P> t	VIF
(Intercept)	608.6468	95.622	6.365	< 0.000001***	/
OfficeArea	0.0581	0.014	4.122	0.00001***	1.00
NumBldgs	1.0322	0.272	3.795	0.00001***	1.24
NumFloors	18.9284	1.148	16.489	< 0.000001***	2.36
BuiltFAR	86.1884	2.395	35.985	< 0.000001***	2.31
ResidFAR	20.1937	2.811	7.183	< 0.000001***	2.96
CommFAR	27.4193	2.591	10.583	< 0.000001***	1.62

Term	Estimate	Std. error	Statistic	P> t	VIF
BoroCode	-23.7046	4.077	-5.814	< 0.000001***	2.16
CondoNo	-0.0051	0.002	-2.789	0.005***	1.19
land_use	-126.8431	7.821	-16.218	< 0.000001***	1.45
Built_year	-0.2815	0.064	-4.369	< 0.000001***	1.35
n_theatre	8.6670	2.592	3.344	0.0007***	1.53
n_restaura	2.2629	0.131	17.292	0.001***	3.33
n_bank	3.5187	0.996	3.533	< 0.000001***	1.89
HubDist	-0.0028	0.001	-3.131	< 0.000001***	3.15
shoreline	19.4067	5.563	3.489	0.002***	1.15
NO	-6.8586	0.899	-7.627	< 0.000001***	2.14
O ₃	-14.8841	2.993	-4.972	< 0.0000001***	3.01
N			5025		
R ²			0.714		
R ² adjust			0.713		

Note: ***: Significant at 99% confidence level. **: Significant at 95% confidence level

Table 3 presents the global Moran's I of land values and O₃ and NO concentration. The value of Moran's I for land values is 0.578, while for O₃ and NO, the figures are 0.880 and 0.653, respectively. These results indicate a significant and positive spatial autocorrelation for land values, O₃ and NO concentration. This observation aligns with findings from previous studies (Wang et al., 2017a; Wang et al., 2022b).

Table 3. Global Moran's I for land values, O₃ and NO concentration

	Moran's I	z-score	p-value
Land values	0.578	150.933	0.001
O ₃	0.880	226.185	0.001
NO	0.653	167.539	0.001

Positive Moran's I indicate significant spatial dependence of land values and air pollution at the micro-level. Table 4 presents the results of SLM and SEM models estimating the relationships between land values and O₃, NO, showing that the spatial autocorrelation parameters are significant with positive coefficients in both models, which aligns with our expectations. The spatial regression models exhibit superior performance compared to their non-spatial counterparts, evident in higher R², and the SEMs slightly outperform SLMs. The results consistently confirm

the negative impact of O₃ and NO on land values, with coefficients of -15.992 and -5.775 in SEM, respectively.

Table 4. Estimation of spatial regression models

Variable	SLM			SEM		
	Estimate	Std. error	Statistic	Estimate	Std. error	Statistic
O ₃	-9.096***	2.801	-3.247	-15.992***	4.231	-3.779
NO	-5.029***	0.840	-5.981	-5.775***	1.281	-4.507
ρ/λ	0.364***	0.015	23.975	0.449***	0.018	24.385
Control other variables	Yes			Yes		
R ²	0.750			0.751		

Note: ***: Significant at 99% confidence level.

Additionally, we used the residual distribution plot to test the linear assumption of the regression model. Results showed that the linear assumption is dissatisfied, which demonstrated the necessity of using machine learning algorithms to reveal the non-linear relationship between explanatory variables and the prediction variable.

5.2. Performance of machine learning models

We employed five machine learning algorithms for comparison with the OLS model and used cross-validations to stabilize the outcomes. Cross-validation is a standard practice in literature and is often recommended for balancing the trade-off between computational efficiency and robust model evaluation. However, there is a risk of overfitting in cases where the model is complex and data is limited. The model might perform well within the folds but struggle with new, unseen data. We selected 5-fold cross-validation as recommended in previous studies (Ma et al., 2020; Hu et al., 2022; Jin et al., 2022), which allowed us to obtain robust performance estimates without excessively taxing computational resources. With a sample size of 5025, our dataset provides ample observations to ensure that each fold in the 5-fold cross-validation process sufficiently represents the overall population.

The results are summarized in Figure 3 and Table 5. Overall, most of the algorithms presented good performance, with R^2 in the range of 0.512 to 0.863. The MAE values range from 0.299 to 0.605, the MSE ranges from 0.166 to 0.592, and the RMSE ranges from 0.407 to 0.769. GBDT, RF, and XGBoost perform better compared with OLS. Among the 5 machine learning techniques, GBDT, RF, and XGBoost showed minor differences in prediction errors. The MAE of GBDT and RF are around 0.304 and 0.307, compared to 0.299 for XGBoost. Similarly, the MSE and RMSE of GBDT and RF are slightly higher than their XGBoost counterparts. While the R^2 for XGBoost is 0.863, the figures for GBDT and RF are 0.857 and 0.852, respectively. The R^2 of XGBoost in 5-fold cross-validation is 0.852, which reveals that overfitting is less likely to have occurred in the result. These results show that XGBoost, a non-linear machine learning method, improves the model accuracy and performs better than OLS, SLM, SEM, KNN, ETR, GBDT, and RF. Therefore, XGBoost is used to identify the relative importance of pollutant variables. Although we only explored the impact of air pollutants on residential lots by using 17 variables, the performance of our models is relatively good compared with the study of Ma et al. (2020), indicating the reliability of our prediction results.

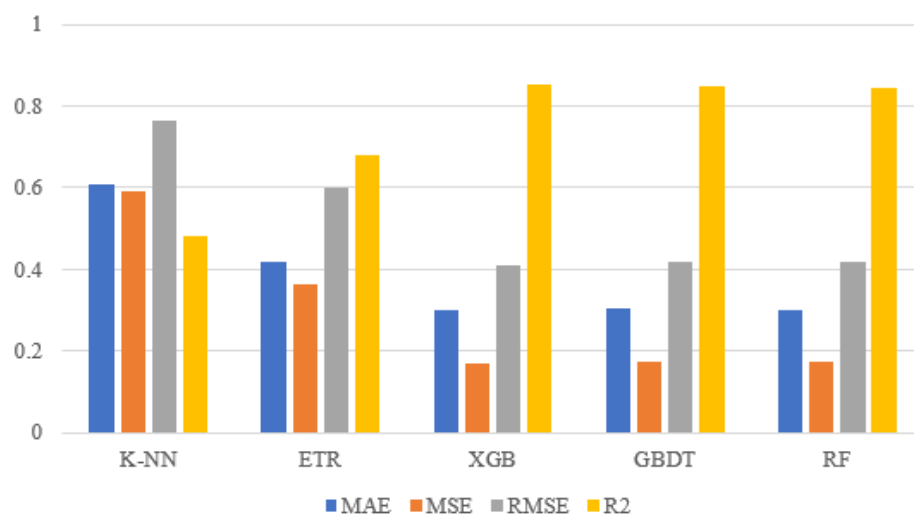


Figure 3. The 5-fold cross-validation performance of 5 machine learning models

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Table 5. Individual machine learning performance

Model	MAE	MAE (CV)	MSE	MSE (CV)	RMSE	RMSE (CV)	R ²	R ² (CV)
K-NN	0.6058	0.6095	0.5928	0.5911	0.7699	0.7654	0.5125	0.4805
ETR	0.4293	0.4184	0.3603	0.3626	0.6002	0.6017	0.7037	0.6805
XGB	0.2993	0.3002	0.1660	0.1675	0.4074	0.4092	0.8635	0.8526
GBDT	0.3041	0.3021	0.1735	0.1731	0.4165	0.4160	0.8573	0.8477
RF	0.3079	0.3013	0.1797	0.1753	0.4239	0.4185	0.8522	0.8458

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Note: MAE, MSE, RMSE, R² are calculated on the validation set.

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MAE (CV), MSE (CV), RMSE (CV), R² (CV) are average metrics from 5-fold cross-validation.

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5.3. The importance of explanatory variables

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Figure 4a shows the ranking of the explanatory variables based on their importance.

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Overall, it is evident that the built environment and lots' characteristics are the two most significant

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contributors to land values. Notably, the built area ratio is the most dominant contributor, which

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aligns with the study of Hu et al. (2016), accounting for nearly 35.27% of the impact on land

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values. This is logical because FAR illustrates the connection between the overall usable floor area

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of a building and the total lot area on which the building is situated (Kok et al., 2014). A greater

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FAR suggests a densely populated or highly urbanized area, leading to an increased land value

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(Ma et al., 2020). The second and third largest contributors are the number of floors and

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restaurants, implying that a higher number of floors and restaurants corresponds to higher land

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values. The percentages for these factors are 14.73% and 13.91%, respectively. It is reasonable to

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expect that areas with a greater concentration of POI could exhibit elevated land values, as a higher

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POI density signifies increased human activity intensity (He et al., 2018), which is indicative of

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advanced development levels (Capozza and Helsley, 2017). The borough where lots are located

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and the distance to the city center significantly contribute to land values (6.55% and 11.87%,

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respectively). These findings are consistent with previous studies on property prices (Heikkila et

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al., 1989; Xiao et al., 2019). The year of building and residential floor area ratio are also essential

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to land value prediction (2.62% and 2.86%). The three least important variables are office area,

commercial floor area ratio, and the number of theatres within a 500-meter radius, accounting for under 1% each.

Regarding the pollution variables, both O₃ and NO are significant explanatory pollutant variables. The total contributions of pollutant variables to land values are 3.11% (see Table 6). Among 17 explanatory variables, O₃ ranks as the eleventh most important variable to land values, with a 1.64% contribution. Similarly, NO ranks 12th with a 1.47% contribution. Notably, the contributions of NO and O₃ are higher than certain building environment features, such as the commercial floor area ratio, number of banks, number of theatres, and office areas. Figure 4b visually illustrates how each variable contributes to land values. Each data point represents a lot. Plots with higher O₃ and NO concentrations (red dots) have lower SHAP values than those with lower air pollutants (blue dots). It means that the concentrations of O₃ and NO negatively impact land values.

To sum up, the contribution of variables can be briefed as follows: (1) Built environment factors are the most important in the valuation of land values; (2) Although the impact of air pollution on land values accounts for a minor value, air pollution variables such as O₃ and NO are more important than some built environment variables, including the commercial floor area ratio, number of hospital number of banks, number of theatres, and office area. Thus, the influence of pollutants on land values should not be ignored and needs to be accounted for different valuation purposes.

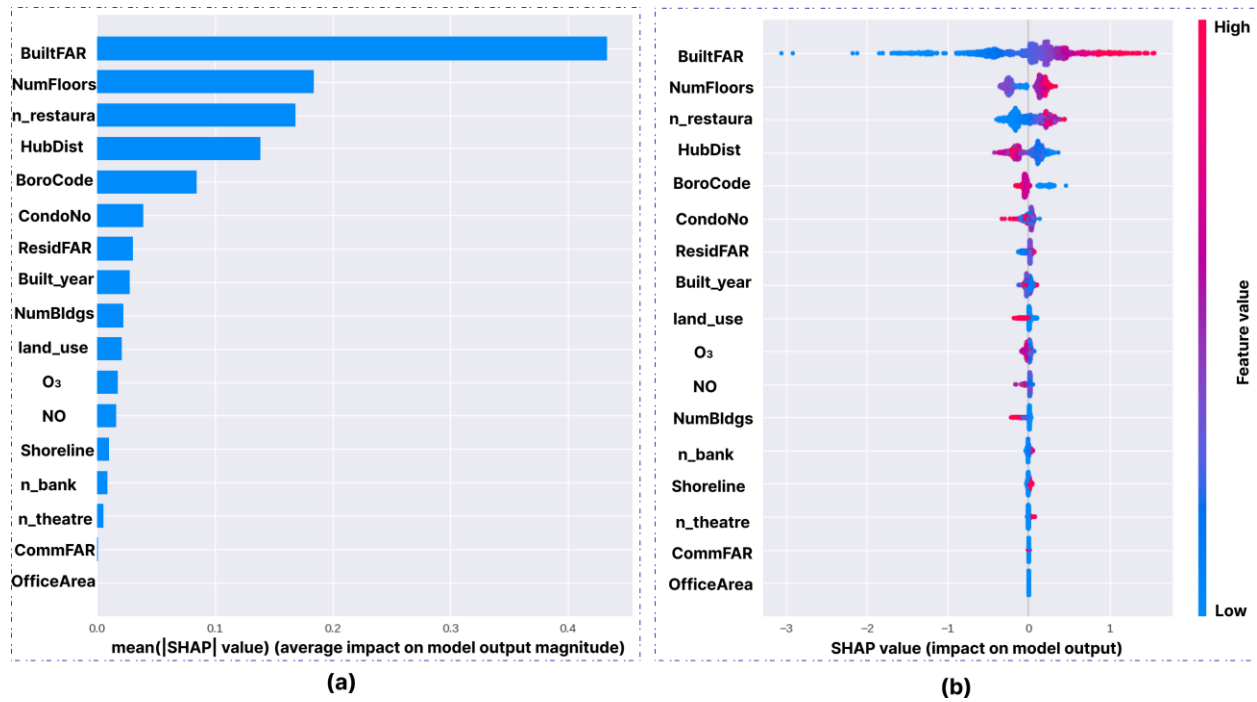


Figure 4. SHAP analysis results by land values

Table 6. The relative importance and ranking of explanatory variables.

Variables	Mean (SHAP value)	% Contribution	Important level
Air pollution variables			
Ozone	0.020	1.64 %	11
Nitric Oxide	0.018	1.47 %	12
Sum	0.038	3.11 %	-
Remaining variables			
BuiltFAR	0.431	35.27 %	1
NumFloors	0.180	14.73 %	2
n_restaura	0.170	13.91 %	3
HubDist	0.145	11.87 %	4
BoroCode	0.080	6.55 %	5
CodoNo	0.046	3.76 %	6
ResidFAR	0.035	2.86 %	7
Built_year	0.032	2.62 %	8
NumBldgs	0.023	1.88 %	9
Land_use	0.021	1.72 %	10
Sum of 5 other features	0.021	1.72 %	-
Sum	1.222	96.89 %	-

5.4. Nonlinear effects of explanatory variables

We have selected air pollution variables and other variables as the top 9 important features. These variables used for non-linear exploration encompass the built floor area ratio, number of restaurants, number of floors, distance to the city center, BoroCode, number of condominiums, year built, number of buildings, residential floor area ratio, O₃ concentration, and NO concentration.

Figure 5 illustrates the non-linear relationship between explanatory variables and land values, emphasizing complex associations. The built area ratio, the number of floors and restaurants, positively influence land values (Figure 5 a, b, c). When the built area ratio exceeds 9, its impact on land values remains stable. It is noticeable that land lots with a BuiltFAR higher than 9 and a higher number of restaurants tend to have greater values. The land values increase rapidly as the number of floors increases, but only up to five floors (Fig 5b). Similarly, the presence of restaurants shows a positive association with land values; that is, as the number of restaurants within 500m increases, the land values increase. However, beyond a certain threshold (40 restaurants), the influence on land values reaches a plateau, even with a further decrease in the distance to the city center (Fig 5c). In contrast, the distance to the city center (Fig 5d) and the number of condominiums negatively impact land values. Specifically, the land values show minor differences between a radius of 3000 and 8000 meters from the city center. However, beyond this threshold, the influence on land values shifts from stable to negative. From 8000 meters to 10000 meters, the prices decrease rapidly before a slight reduction thereafter. The non-linear effect with BoroCode is shown in Fig 5e. Similarly, the Shapley values corresponding to the number of condominiums depict a non-linear relationship, as shown in Fig 5f. The Shapley value exhibits a slight growth before decreasing at the threshold of 3000. Moreover, the residential floor area ratio positively influences land values (Fig 5g). However, a subsequent increase in the residential floor

area ratio (exceed 4) is associated with a trivial increase in land values. The non-linear relationship between construction years and land values was also evident (Fig 5h), showing the negative association as the construction year increases from 12 to 25 years. The land values decrease when the construction year exceeds 25 and below 125 years. Plots with ancient buildings constructed more than 125 years ago tend to have higher values.

Interestingly, plots with older residential buildings (>100 years) with elevators exhibit lower value than walk-up buildings of the same age. We hypothesized that elevators in older buildings could be components of ageing infrastructure, making them more susceptible to breakdowns and malfunctions. This could lead to a perception of inconvenience and a potential decrease in property value. Besides, cities have turned increasingly to historic preservation of residential property. Historic preservation, however, imposes strict rules that make property improvement more expensive than it otherwise would be and positively impact property values (Clark and Herrin, 1997; Been et al., 2016). The number of buildings has a negative association with land values that gradually decrease as the number of buildings increases (Fig 5k). Lots with a high number of buildings and a lower built-area ratio have lower values than lots with low buildings and a high built-area ratio.

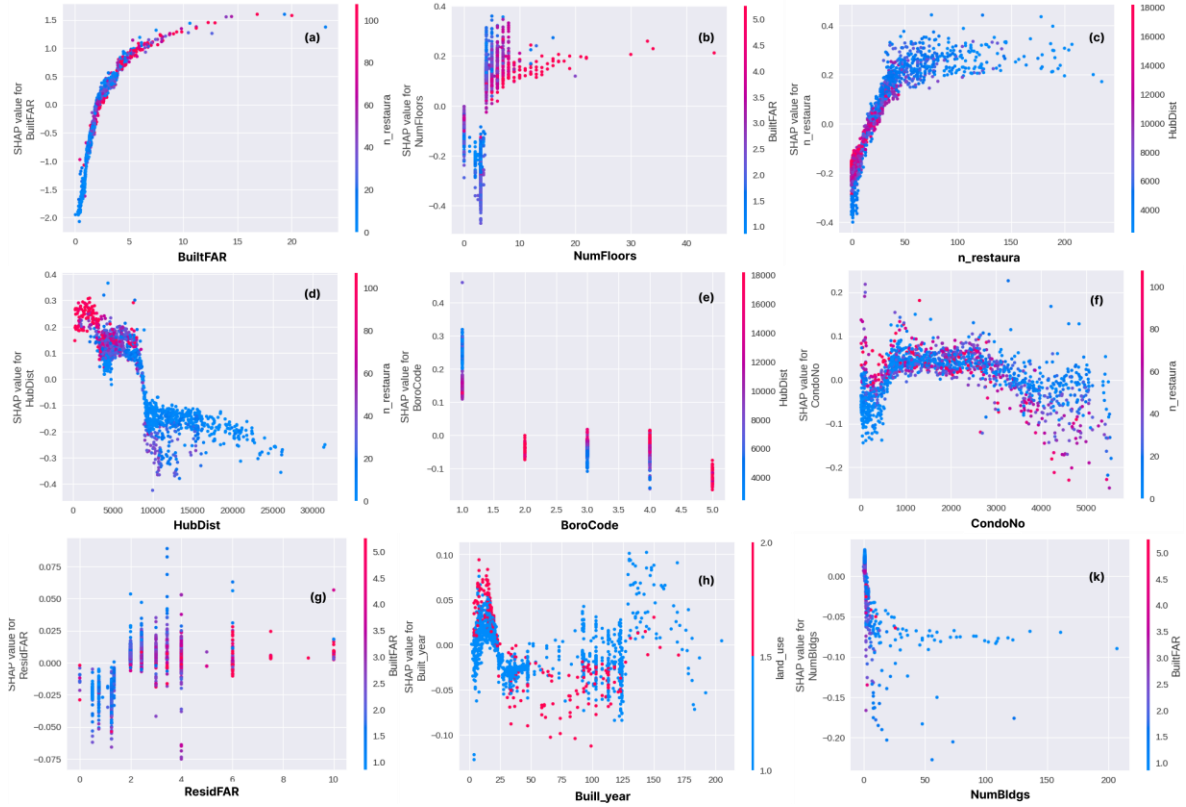


Figure 5. Non-linear relationships between the built environment variables and land values

We found a non-linear relationship between pollutant variables (O_3 and NO) and land values. Overall, the impacts of pollutants on land values are negative (Fig 6). When O_3 concentration is below 29.3 ppb ($\sim 57.52 \mu g/m^3$), which is the average O_3 concentration of lots in New York, the influence of O_3 concentration on price is negligible. However, the influence gradually increases when the concentration is higher than the average. As for NO , at a concentration of 13 ppb, its effect on land values changes from positive to negative. This indicates that land values decrease when the NO concentration exceeds 13 ppb. Interestingly, there is a positive influence of NO on land values when NO concentration is at a low level, probably due to the low NO concentration in suburban areas where land values also tend to be low. The insignificant impact of low NO concentration on human health made local residents less sensitive to it and resulted in a positive effect. The impact of NO on human health becomes more precise; people are more concerned with accommodation decisions.

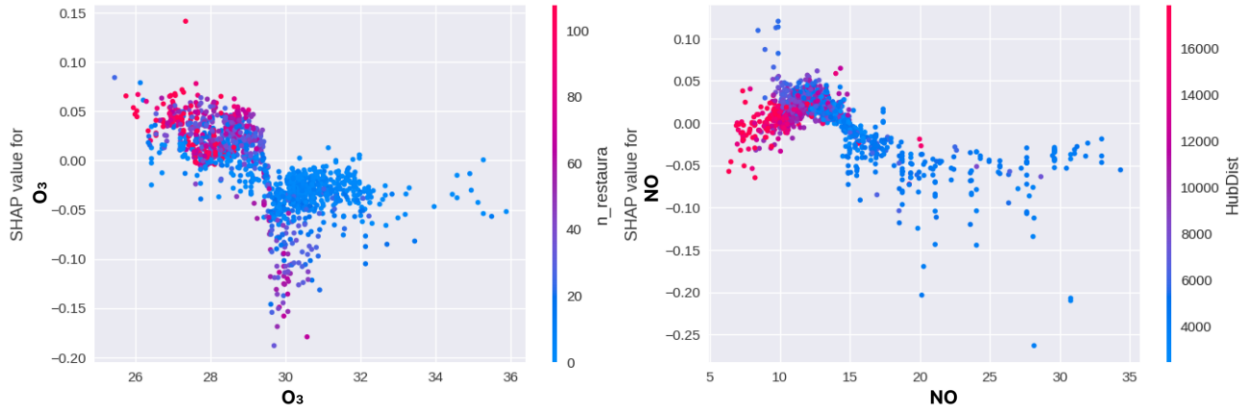


Figure 6. Non-linear relationships between O_3 , NO and land values

In figure 7, we can observe the local explanations for five randomly selected cases from each borough. For case number 1 in the Bronx, the variables BuiltFAR and NumFloors have a positive impact (red color), while n_restaurant, HubDist, and Condono have a negative effect (blue colour). Additionally, O_3 negatively influences the lot's value, with a concentration of 29.56 ppb. Similarly, O_3 has a negative effect on the lot's price in the second case in Queens and the fourth case in Brooklyn, but the impacts of NO in 5 cases are minor. In the third case in Manhattan, O_3 and NO have infinitesimal influences on the land values. The surrounding environment of case 3 is characterized by a dense of urban facilities that positively impact land values. Interestingly, air pollution variables had almost no effect on the land values in the fifth case (Staten Island). We identified six key clusters (zones A, B, C, D, E, F) characterized by O_3 and two clusters (zones G and H) characterized by NO concentrations exceeding the established threshold. In response to this finding, targeted pollution interventions are recommended for these zones, alleviating NO and O_3 pollution's adverse effects on residential areas and the real estate market.

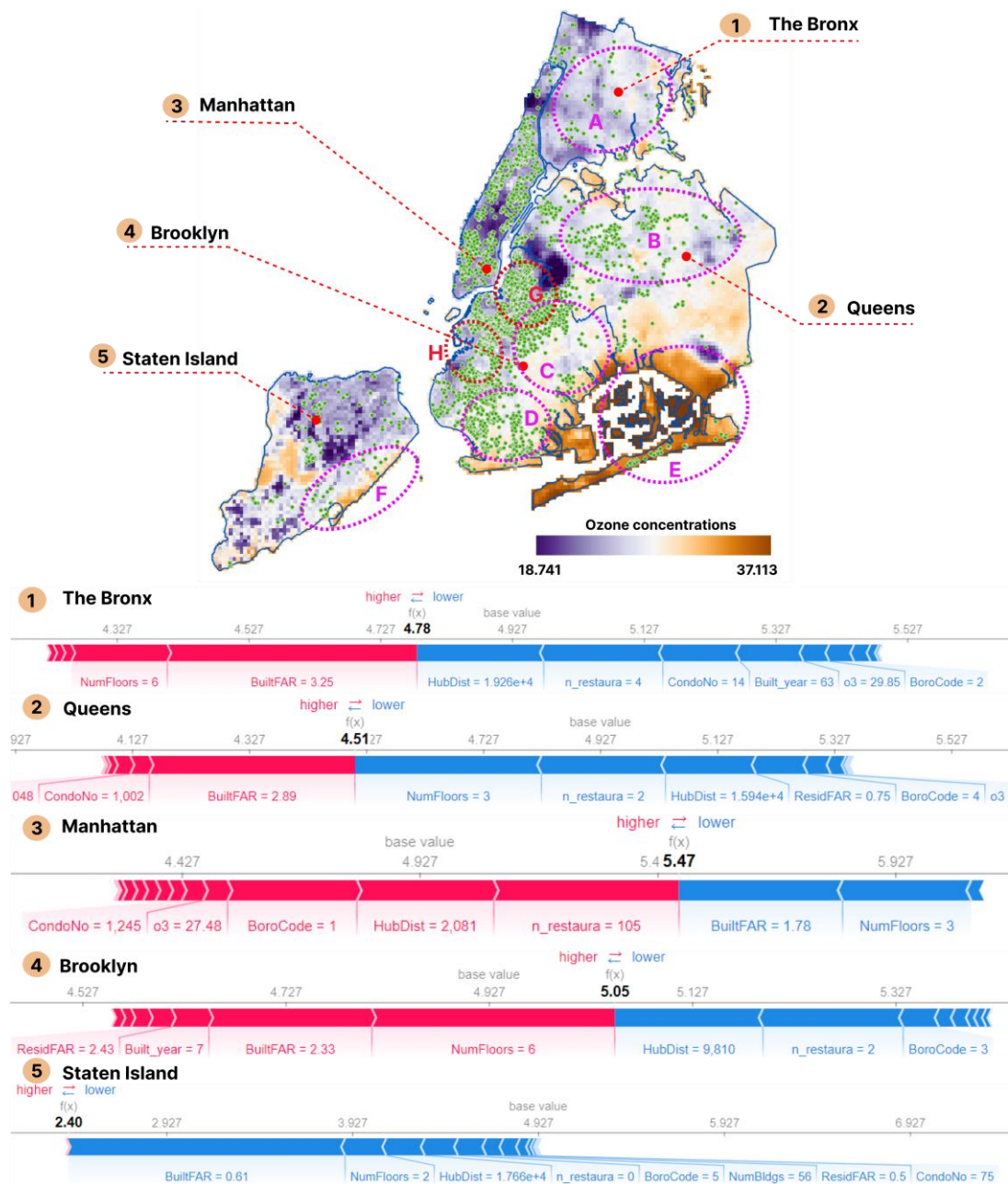


Figure 7. Local explanations for five cases randomly from each borough

6. Discussion

6.1. The nonlinear and threshold effects of air pollution on land values

Consistent with previous studies and our expectations, air pollution has a negative impact on land values. Researchers have identified relationships between property prices and air pollution (Tang and Yiu, 2010; Mínguez et al., 2013; Chen and Chen, 2017; Chen and Jin, 2019; Amini et

al., 2022; Zhang et al., 2023a). They have observed that purchasers are willing to pay a premium to avoid areas with high air pollution, as it affects their willingness to invest in housing. As awareness of the adverse health effects of air pollution increases, individuals become more cautious in housing choices to avoid polluted areas. However, most studies have limitations concerning non-linear relationships and interaction effects between variables. To address these limitations, our study used machine learning algorithms to confirm and analyze the non-linear relationships between air pollution and land values.

We confirmed the presence of non-linear relationships between air pollution variables and land values. Not every increase in pollutant concentration leads to a fall in land values; land values begin to be negatively impacted if pollutant concentration exceeds a certain threshold. These findings corroborate a previous study conducted by (Zou et al., 2022), which identified the non-linear impact of pollution on house prices at the county level. In particular, we found that land values are likely to reduce when the summer average O₃ concentration exceeds 29.3 ppb (57.52 µg/m³), the identified threshold where buyers are willing to pay for lower O₃ pollution. The reduction in land values aligns with the housing decisions. People tend to realize the negative effects of air pollution on health as pollution levels increase, and households may move to cleaner air locations after an adult is diagnosed with asthma (Pan, 2023). Furthermore, studies have shown that the number of hospital admissions for cardiovascular and respiratory diseases increases due to O₃ exposure above 5 ppb (Anbari et al., 2022), and an increase of 13 ppb in O₃ concentration leads to 2.9-8.4% and 5.4 - 6.5% asthma emergencies in boys and girls, respectively (Sheffield et al., 2015). The increase in visits, hospitalizations, and deaths due to ozone exposure (see Table 7) has raised public concerns about its negative effects. As a result, households may be willing to pay more for less pollution (Bayer et al., 2009; Chakraborti et al., 2019) or willing to incur costs

because of avoidance behaviors such as alternative modes of transportation and migration (Enrico and Matthew, 2011; Chen et al., 2022a).

Table 7. Health impact of Ozone concentration (2015-2017)

	Asthma emergency department visits	Annual number of asthma hospitalizations (age 18+)	Annual number of asthma hospitalizations (under age 18)	Cardiac and respiratory deaths
Bronx	962.7	97.9	132.7	67.5
Brooklyn	1057.1	63.4	104.2	131.2
Manhattan	635.7	38.4	45.9	64.9
Queens	515.9	49.4	56.5	111.5
Staten Island	126.6	13.6	12.3	35.7

Source: Environment and Health Data Portal, NYC

Our findings regarding the negative impact of O₃ concentration on land values align with the results of Hitaj et al. (2018), Saptutyningsih and Ma'ruf (2015) and Chakraborti et al. (2019), who identified the negative linear effect of O₃ pollution on property prices. However, our findings contrast with a study conducted by Mei et al. (2020) and Usharani (2021), which revealed positive associations between O₃ and SO₂ with house prices. It is worth noting that the low and standard ranges of SO₂ concentration over the years may explain the presence of a positive relationship in their studies. The low SO₂ concentration over the years made local residents less sensitive to it. The positive association between prices and O₃ may arise from the public insufficient understanding. Additionally, the lack of non-linearity exploration in house prices could lead to the observed positive relationship with O₃ when pollution concentrations are under the mean level, as demonstrated by our study.

We find that NO concentration has a negative influence on land values, which is consistent with previous studies (Chen et al., 2018; Amini et al., 2022). However, we detected that NO concentrations show a positive association with land values when their concentration is below the mean threshold, inconsistent with the findings of Won Kim et al. (2003), who indicated an

insignificant impact of NO_x on house prices. The positive impact of low NO concentrations on land values in our study aligns with the study of Tang and Niemeier (2021), who found a positive relationship between NO and property prices at 12.86ppb in California, USA. The positive association can be explained as follows. First, in urban areas, particularly those undergoing development or with high economic activity, the benefits of the proximity to urban amenities and economic opportunities may outweigh the negative impact of NO (Azmi et al., 2012). For a developed area like NYC, transportation may contribute a larger proportion to the NO emission (NYC Health, 2022). Therefore, areas surrounding the main roads tend to exhibit higher levels of NO. These areas may also have higher land values due to their convenient locations. Therefore, as long as the concentration of NO is not high enough to cause concern, people are willing to pay higher prices for houses with more convenient transportation access (Jiao and Liu, 2010). Second, suburban areas generally exhibit lower NO concentrations. Due to the lower land values in suburban areas compared to central urban areas with low NO concentrations, a positive correlation between NO concentration and land values can be observed in this context. However, the impact changed to negative when the NO concentration exceeded the threshold at the mean value. The threshold effect aligns with the protection motivation theory of Rogers (1975), who proposed a social cognition developed originally to examine dear appraisal and behaviour change. In this context, the theory might not be directly related to environmental economics. We argue that the protection motivation theory is directly relevant to individuals who perceive the risks associated with air pollution and protection actions. For example, if residents believe that their health is vulnerable to the effects of air pollution, it might influence their preferences for certain locations (Chen et al., 2022a) or their willingness to pay for properties in less polluted areas (Liang, 2023).

Furthermore, if residents believe that moving to areas with lower air pollution levels would effectively protect their health, it could impact their settlement decisions (Zhang et al., 2022a).

We comprehensively assessed various pollutants, including PM_{2.5}, as presented in Table 1. Through stepwise regression, PM_{2.5} was excluded from the final model due to its comparatively less significant impact than other pollutants within the study areas. This exclusion aligns with the observed trend of decreasing PM_{2.5} concentrations in recent years. Annual average levels of PM_{2.5} declined by 43% since 2009 (NYC Health, 2022). The prioritization of O₃ over PM_{2.5} in our analysis reflects residents' heightened concerns regarding O₃-related health issues. The sensitivity of the population to O₃ underscores its relevance in our study context. It is essential to emphasize that our established methodology can be adapted to assess the impact of other pollutants. In regions where PM_{2.5} is a predominant pollution source, its significance may be more pronounced, and our approach remains applicable to accommodate such variations.

We also recognised that most built environment variables had a forceful influence on land values, consistent with existing studies (Huang and Yin, 2015; Hu et al., 2016; Li et al., 2021; Qiu et al., 2022; Wang et al., 2022c). However, we also examined the non-linear relationship between built environment characteristics and lots values, which is rarely confirmed by previous studies (Ma et al., 2020).

Our study consistently affirms the robust performance of machine learning models, with XGBoost, RF, and GBDT demonstrating superior accuracy in predicting land values compared to OLS, SLM, and SEM models. Specifically, XGBoost exhibits the highest accuracy in our analyses, a result consistent with the findings of Chen et al. (2023a). In the validated dataset, XGBoost's performance surpasses the OLS model by 19.4%. Traditional linear-based studies propose that increased pollution concentration might diminish land values (Chen et al., 2018; Amini et al.,

2022). However, this linear assumption lacks consideration for threshold effects, an aspect effectively addressed by machine learning and SHAP in our study. Furthermore, machine learning models, equipped with feature importance analysis, empower us to identify the relative significance of variables (Li, 2022), providing nuanced insights valuable for policymakers and urban planners. Furthermore, by employing SHAP, we revealed synergies/antagonistic relationships that the hedonic model can not uncover (figure 5 & 6).

Although air pollution accounts for around 3.11% of the contribution to land values, its impact cannot be avoided. Implementing effective pollutant control policies is essential for several reasons. Firstly, it reduces the implicit losses in the real estate market and supports environmental policies aligned with the green economy initiatives of many governments. Secondly, considering air quality in land values and house prices can encourage builders to comply with environmental quality policies, ensuring adherence to regulations and avoiding potential financial penalties. Thirdly, considering the non-linear correlation between housing prices and air pollution, tailored pollution control strategies need to be formulated based on the distinct air quality conditions observed in various districts. Lastly, we suggest implementing a land value capture mechanism since air pollution can impact land values. This mechanism could serve as a funding source for initiatives to enhance sustainable transportation and housing, contributing to improving urban air quality. In summary, pollution policies are crucial in shaping the real estate and housing sector towards sustainability, public health, and social welfare. By implementing effective pollution control measures, a more attractive and sustainable living and working environment can be created, benefiting property owners, residents, and businesses.

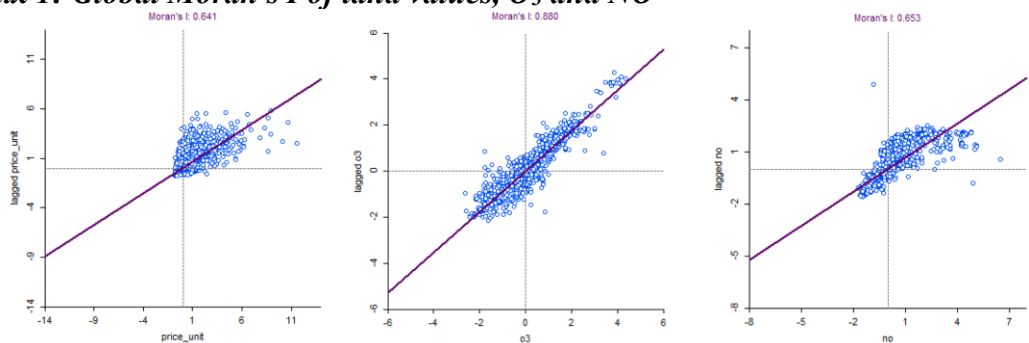
6.2. Limitations and future works

Despite methodological advancements in environmental housing, certain drawbacks still need to be considered. First, only data on land values in NYC for a single year is used in this analysis. The strategy and results for various cities over several years might be validated using additional data and conducting a more thorough investigation. Second, we use single pollutant indicators to provide specific information about individual air pollutants (e.g., ozone, NO) instead of the AQI (air quality index). The mainstream calculation methods use the pollutant with the most severe pollution to characterize the AQI (Cheng et al., 2007). The AQI can better reflect the overall pollution status of a region. Therefore, the AQI results can more accurately reflect the impact of overall air quality on residents' choice of housing locations, as people generally do not consider the concentration of just one pollutant when making decisions. Furthermore, it's also worth mentioning that we observed the concentrations of individual pollutants, allowing us to compare the differences between different pollutants and their causes and identify key pollutants that need to be controlled and their management targets. While O₃ and NO partly explain the AQI index, future work can also include the AQI to obtain more comprehensive conclusions. Besides, we do not include socioeconomic characteristics such as race, income, or job opportunities. Future studies can integrate these characteristics to deeply understand different ethnicities' settlement preferences. Future studies could incorporate rent data by using rent-price ratio analysis. This ratio can indicate the rental yield and affordability of housing in a specific area. Changes in this ratio may reflect the impact of air pollution on the demand for land/housing. Finally, we used land lots as an analysis unit to explain the non-linear relationships between air pollution and land values. In future studies, these relationships could be examined using ecological and individual exposure approaches.

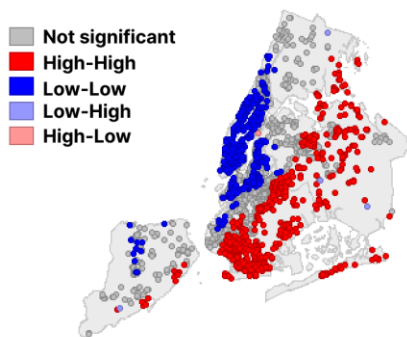
7. Conclusion

Utilizing various data in New York City, USA, this study examines the non-linear relationship between air pollution and house prices at a microscopic scale. Firstly, we employed OLS regression and conducted multicollinearity analysis to select potential variables related to house prices. We also adopt the spatial lag model (SLM) and spatial error model (SEM) to account for the spatial autocorrelation that may exist among the variables. Secondly, we compared the performance of various machine learning algorithms, including GBDT, K-NN, ETR, RF and XGBoost, to identify the best model before analyzing non-linearity. In this study, the XGBoost algorithm demonstrated superior performance in land value predictions compared to other techniques (i.e. 19.4% higher than OLS). Subsequently, we explored the importance rankings of explanatory variables, identifying O₃ and NO as two significant pollutant variables with respective contributions of 1.64% and 1.47% in affecting land values. Finally, utilizing the XGBoost model and SHAP, we examined the non-linear relationships between O₃, NO, the built environment factors, and land values. Our results revealed that air pollution, specifically O₃ and NO, has a negative impact on land values in general. O₃ and NO do not affect land values if their concentrations are below average. However, their effects become negative when concentrations exceed the regional average. Given the non-linear association between housing prices and air pollution, it is essential to develop customized pollution control strategies that account for the air quality conditions observed in different districts. Since air pollution can influence land values, the land value capture mechanism can be applied to finance sustainable public projects. The contribution of this study is threefold: providing evidence of the non-linear relationship between air pollution and land values, ranking the relative importance of air pollution variables for land values, and serving as a valuable scientific reference for informing environmental regulation implementation and settlement decisions.

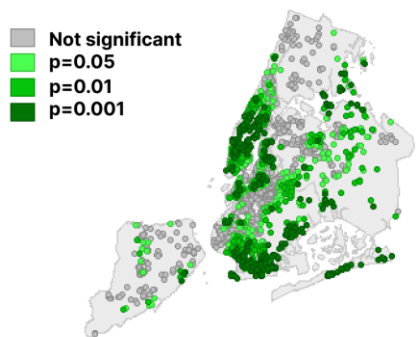
Appendix 1: Global Moran's I of land values, O₃ and NO



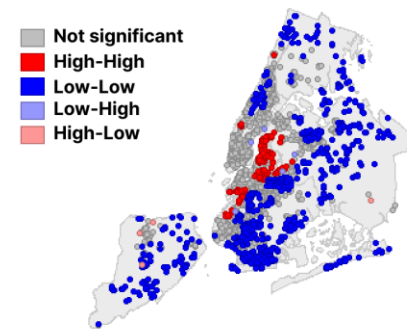
Appendix 2: LISA cluster and LISA significant maps of land values and pollution



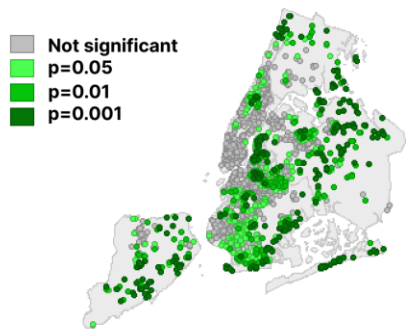
LISA cluster map of O₃



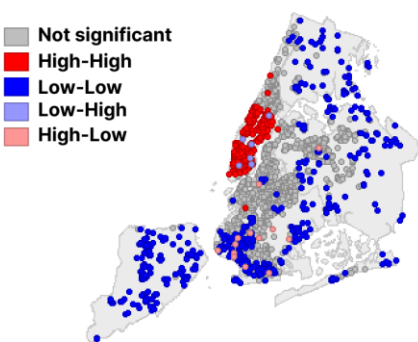
LISA significance map of O₃



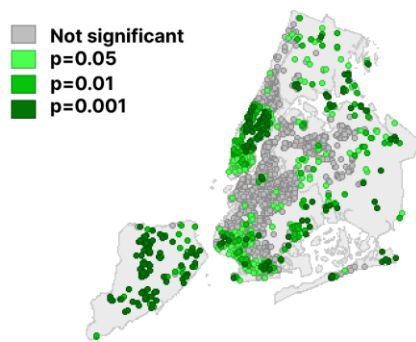
LISA cluster map of NO



LISA significance map of NO



LISA cluster map of land values



LISA significance map of land values

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